Operation Management

BITS Pilani
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<MBACCZG526 (BC-26)Operation Management>
Lecture No. 1

Heritage of Operation Management

Dates	Era	Concepts	Contributor
1769-1880	Industrial Revolution	Steam Engine, Division of Labour, Standardized Parts	James Watt. Adam Smith. Bill Whitney.
1880-1910	Scientific Management	Time and Motion Studies, Process Analysis. Gantt Chart. Queuing Theory.	Frank Gilbert. Francis Taylor. Henry Gantt. Erlang.
1910-1980	Mass Production	Moving Assembly Line. Statistical Sampling	Ford & Sorensen Walter Shewhart.

Heritage of Operation Management

Dates	Era	Concepts	Contributor
1910-1980	Mass Production	EOQ PERT/CPM MRP	Harris. DuPont.
1930-1960	Human Relations	Motivation theories	Elton Mayo. Abraham Maslow. Fredrick Herzberg.
1947-1980	Operation Research	Linear Programming, Simulation, Decision making.	George Dantaig. Remington Rand. Joseph Orlicky.

Heritage of Operation Management

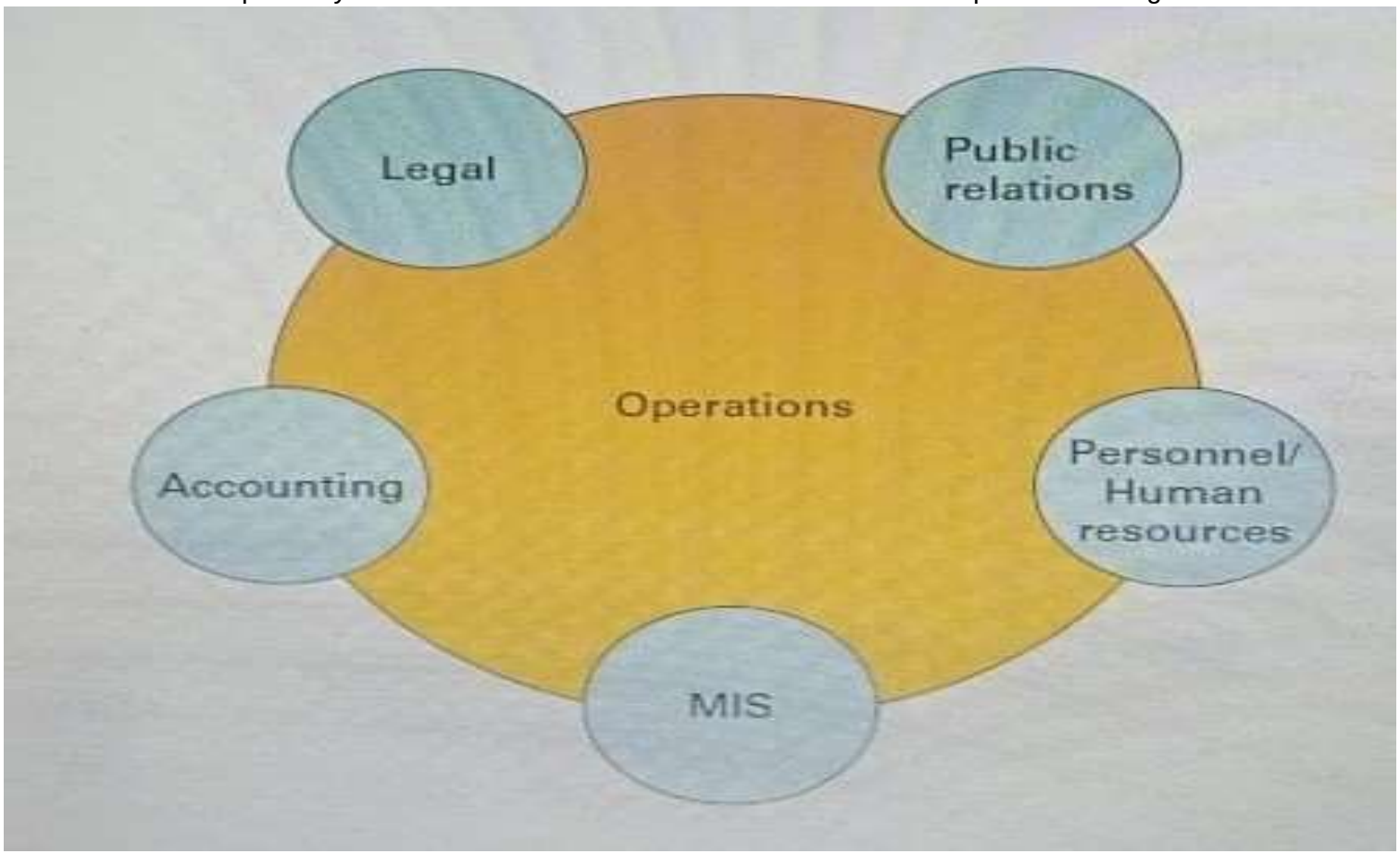
Dates	Era	Concepts	Contributor
1980-1995	Quality Revolution (Lean Production)	JIT, TQM, Six Sigma, CAD, Baldridge award, Kanban	Edward Deming. Joseph Juran. Taichi Ohno. Bill Smith. Philip Crossby.
1995-2005	Mass Customization	Internet/E-commerce, ERP, ISO, Finite Scheduling, SCM, Mass Customization, Build to Order, RFID	SAP. Oracle. Dell Google. Amazon. Yahoo.

Heritage of Operation Management

Dates	Era	Concepts	Contributor
2005-2020	Globalization	WTO, Global Supply Chain, Outsourcing. Growth of Transactional Organization, Instant communication, Logistics Work Ethics	China and India
Present	Industry 4.0 & Sustainable Operation	Circular Economy, IOT, IIOT, Drones	Numerous Scientist

Why Operation Management

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Operation Management by William J Stevenson

Why Operation Management

- How people organize themselves for productive enterprise.
- The role of Operation managers.
- Optimize the resources as Operation Management is a costly affair.
- How goods and services are produced.

Organizing to Produce Goods and Services

1. Marketing, which generates the demand, or at least takes the order for a product or service.
2. Production / Operation, which creates, produces and delivers the product.
3. Finance/ Accounting, which tracks how well the organization is doing, pays the bill, and collects the money.

Manufacturing Unit

Operation

- Construction, Maintenence.
- Production & Inventory Control, Scheduling, Materials control.
- Quality Assurance and control.
- Supply Chain Management.
- Tooling, Lubrication & Assembly.
- New Product Development and Design.
- Process Analysis.
- Engineering Design.

Finance

- Accounts Receivable.
- Accounts Payable.
- Disbursement.
- Credit.
- Funds Management.
- Money Market.
- International Exchange.
- Capital Requirements.
- Stock Issue.
- Bond Issue and Recall.

Marketing

- Sales Promotion.
- Advertising.
- Market Research.
- Sales.

Career in Operation Management

- Production Manager.
- Quality Manager.
- Supply Chain Manager.
- Process Improvement Consultant.
- Product Manager.
- Project Manager.

Certifications

- Production Manager : WCM, Design Thinking, Operation / Manufacturing Strategy
- Quality Manager : Lean Six Sigma, IMS
- Supply Chain Manager : CPIM, CSCP, CLTD
- Process Improvement Consultant : Six Sigma MBB, BPR, Business Analytics.
- Product Manager : Product Management
- Project Manager : PMP, Prince2

Ethics in Operation Management

- Report accurate financial conditions.
- Look into worker safety. Provide them adequate training.
- Provide safety to the product and equipments.
- Give utmost importance to Quality.
- Do not let your operation to harm the community and environment.
- Promise long term jobs (avoid hiring and firing principle).
- Respect Worker's rights.

Operation Management

- The management of systems or processes that produce goods and/or provide services.
- Goods: Physical items produced by any organization that has value.
- Services: Activities that provide a combination of time, location, form, and psychological value.
- At enterprise level the production of goods and provision of service is measured in terms of productivity.

Good Vs Service Operation

Characteristics	Goods	Services
Output	Tangible	Intangible
Customer contact	Low	High
Labour	High	Low
Customization	Low	High
Uniformity	High	Low
Inventory	High	Low
Wages	Narrow range	Wide Range
Patentable	Usual	Not Usual

The Service Process Matrix

Degree of Labour Intensity	Degree of Interaction & Customization: Low	Degree of Interaction & Customization: High
Low	Service Factory: Airlines, Transportation, Hotels	Service Shop: Hospitals, Repair services.
High	Mass Service: Retailing, Schools, Banking, Whole selling	Professional Service: Doctors, Lawyers, Accountants, Architects.

Value

- Value is the worth of the net product, personnel, service, and social **benefits a customer receives** in **exchange for the total cost** for a market offering.

Value = Total Customer Benefits – Total Customer Cost.

Total Customer Benefits = Product Benefit + Service Benefit + Personnel Benefit + Social Benefit.

Total Customer Cost = Price + Time Cost + Energy Cost + Psychological Cost.

Any Question

Class Discussion

Identify the Cost (may be more than one) and Benefit (may be more than one) associated by the following.

- Maruti Suzuki Cars.
- Harley Davidson Bike.
- Apple.
- Walmart.
- Netflix.

Production System

Input	Process	Output
Material	Cutting, Drilling	Goods
Machines	Teaching	Services
Labour	Transportation	
Capital	Farming	
Facilities	Debugging	
Energy	Analysing	
	Searching	

Productivity

- **Productivity** is an index that measures output (goods and services) relative to the input (labor, materials, energy, and other resources) used to produce it.
- It is usually expressed as the ratio of output to input.
- $\text{Productivity} = \dfrac{\text{Output}}{\text{Input}}$
- $\text{Productivity Growth} = \dfrac{\text{Current Productivity} - \text{Previous Productivity}}{\text{Previous Productivity}} * 100$

Types of Productivity

Type of Productivity	Characteristics
Labour Productivity	• Units of Output per labour hour. • Units of Output per shift. • Value-added per labour hour. • Dollar value of output per labour hour.
Machine Productivity	• Units of Output per machine hour. • Dollar value of output per machine hour.
Capital Productivity	• Units of Output per dollar input • Dollar value of output per Dollar input.
Energy Productivity	• Units of Output per KWh • Dollar value of output per KWh.
Material Productivity	• Units of Output per unit of material consumed.

Single Factor & Multifactor Productivity

- Multifactor Productivity =

$$\frac{\text{Output}}{\text{Labour+Machine+Capital+Energy+Material+Miscellaneous}}$$

- Single factor Productivity = $\frac{\text{Output}}{\text{Any one factor}}$

- $\frac{\text{Output}}{\text{Labour}}$ = Labour Productivity. $\frac{\text{Output}}{Material}$ = Material Productivity

- $\frac{\text{Output}}{\text{Machine}}$ = Machine Productivity.

- $\frac{\text{Output}}{\text{Capit}al}$ = Capital Productivity

- $\frac{\text{Output}}{\text{Energy}}$ = Energy Productivity

Factors Affecting Productivity

- Productivity depends on Labour, Capital and Management.
- Some other factors like Quality, Technology and Methods play crucial role.
- Process Yield is a terminology that is often used in industry to measure productivity.

Any Question

Class Exercise

- A company that processes fruits and vegetables can produce 400 cases of canned peaches in half hour with four workers. What is labor productivity?

- A wrapping-paper company produced 2,000 rolls of paper one day. Labor cost was $160, material cost was $50, and overhead was $320. Determine the multifactor productivity.

Class Exercise

- Determine the productivity for these cases:

a. Four workers installed 720 square yards of carpeting in eight hours.

b. A machine produced 70 pieces in two hours. However, two pieces were unusable.

Class Exercise

- Determine the multifactor productivity for the combined input of labor and machine time using the following data:

i. Output: 7,040 units

ii. Input

iii. Labor: $1,000

iv. Materials: $520

v. Overhead: $2,000

Class Exercise

- Compute the multifactor productivity measure for an eight-hour day in which the usable output was 300 units, produced by three workers who used 600 pounds of materials. Workers have an hourly wage of $20, and material cost is $1 per pound. Overhead is 1.5 times labor cost.

Take Home Exercise

- A health club has two employees who work on lead generation. Each employee works 40 hours a week, and is paid $20 an hour. Each employee identifies an average of 400 possible leads a week from a list of 8,000 names. Approximately 10 percent of the leads become members and pay a onetime fee of $100. Material costs are $130 per week, and overhead costs are $1,000 per week. Calculate the multifactor productivity for this operation in fees generated per dollar of input.

Take Home Exercise

- Productivity can be measured in a variety of ways, such as by labor, capital, energy, material usage, and so on. At Modern Lumber, Inc., Art Binley, president and producer of apple crates sold to growers, has been able, with his current equipment, to produce 240 crates per 100 logs. He currently purchases 100 logs per day, and each log requires 3 labor-hours to process. He believes that he can hire a professional buyer who can buy a better-quality log at the same cost. If this is the case, he can increase his production to 260 crates per 100 logs. His labor-hours will increase by 8 hours per day.

- What will be the impact on productivity (measured in crates per labor-hour) if the buyer is hired?

Take Home Exercise

- Art Binley has decided to look at his productivity from a multifactor (total factor productivity) perspective (refer to previous problem). To do so, he has determined his labor, capital, energy, and material usage and decided to use dollars as the common denominator. His total labor-hours are now 300 per day and will increase to 308 per day. His capital and energy costs will remain constant at $350 and $150 per day, respectively. Material costs for the 100 logs per day are $1,000 and will remain the same. Because he pays an average of $10 per hour (with fringes), determines his increase in productivity.

Any Question

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 1
2. Operation Management by William J Stevenson—Chapter 1 & 2
3. Operations Management by Russell and Taylor—Chapter 1
4. Service Management by Fitzsimmons—Chapter 1

THANK YOU